

Chapter: 03

Problem-solving Agent: A problem solving agent is a type of goal-based intelligent agent that decides what to do by searching for a sequence of actions that will lead to a desire goal state from a known state.

① Goal Formulation: based on the current situation and the agent's performance measure, is the first step in problem solving. Decide what the agent wants to achieve.

② Problem Formulation: is the process of deciding what actions and states to consider, given a goal.

③ Search: —

④ Execute (carry out the chosen plan)

⑤ Measuring Problem-solving performance:

- Completeness:- Is the algorithm guaranteed to find a solution when there is one.
- Optimality:- Does the strategy find the optimal solution.
- Time Complexity:- How long does it take to find solution.
- Space complexity: How much memory is needed to perform the search.

☐ Uninformed Search.

☐ Breadth-first search:

A simple strategy in which the root node is expanded first, then all the successors of the root node are expanded next, then their successors, and so on.

Define

- Initial state
- Action
- Transition model
- Goal test
- Path cost

Search :- Explore the state space using a search strategy.

• ⑤ Uniform-Cost Search:

Uniform-cost search expands the node n with the lowest path cost $g(n)$. This is done by storing the frontier as a priority queue ordered by g .

⑥ Depth-First Search:

Depth-first search always expands the deepest node in the current frontier of the search tree. DFS uses a LIFO queue. A LIFO queue ensures that the most recently generated node is chosen for expansion.

⑦ Depth-limited Search:

Node at depth l are treated as if they have no successors. This approach is called depth-limited search.

- Optimality: Does the algorithm find the optimal solution when there is one?
- Time Complexity: How long does it take to find a solution?
- Space Complexity: How much memory is needed to perform the search?

⑧ Iterative deepening DFS:

Iterative deepening search is a general strategy often used in combination with ~~DFS~~ Depth-First tree search, that find the best depth limit.

⑨ Bidirectional search:

The idea behind bidirectional search is to run two simultaneous searches - one forward from the initial and the other backward from the goal.

Comparing uninformed search strategies:

Criterion	BFS	Uniform Cost	DFS	Depth-Limited	Iterative Deepening	Bidirectional
Complete?	Yes	Yes	No	No	Yes	Yes
Time	$O(b^d)$	$O(b^{1+\lceil \epsilon/b \rceil})$	$O(b^m)$	$O(b^l)$	$O(b^d)$	$O(b^{d/2})$
Space	$O(b^d)$	$O(b^{1+\lceil \epsilon/b \rceil})$	$O(b^m)$	$O(b^l)$	$O(b^d)$	$O(b^{d/2})$
Optimal	Yes	Yes	No	No	Yes	Yes

Informed search:

① Greedy best-first search:

It tries to expand the node that is closest to the goal, on the grounds that this is likely to lead to a solution quickly.

It evaluates nodes by using the heuristic function;

$$f(n) = h(n)$$

~~xxx~~ Iterative deepening search may seem wasteful because states are generated multiple times. It turns out this is not too costly. The reason is that in a search tree with the branching factor at each level, most of the nodes are in the bottom level. In an iterative deepening search, the nodes on the bottom level (d) are generated once, those on the next bottom level are generated twice, and so on, up to the children of the root.

$$N(DFS) = (d)b + (d-1)b^2 + \dots + (1)b^d$$

For example: if $b=10$ and $d=5$, the number are:

$$N(DFS) = 50 + 400 + 3,000 + 20,000 + 100,000 = 123,450$$

$$N(BFS) = 10 + 100 + 1,000 + 10,000 + 100,000 = 111,110$$

So,

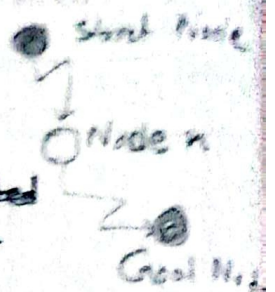
Iterative deepening is the preferred uninformed search method when the search space is large and the depth of the solution is not known.

3.5.2 A* Search

$$f(n) = g(n) + h(n)$$

the cost to reach the node from start node

Heuristic function: estimate cost to get from the node to the goal.



4/ Heuristic Function and its application:

In fig-3.2 and 3.22

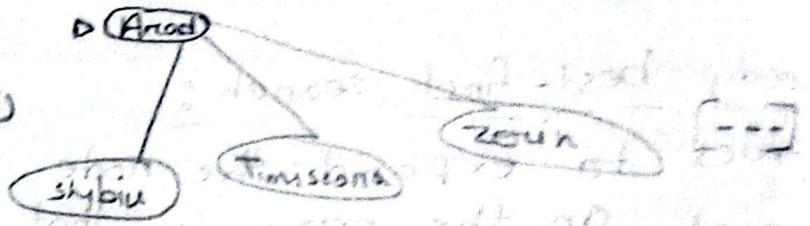
Frung List: [Anad]

(a) The initial state

$$f(\text{Anad}) = g(\text{Anad}) + h(\text{Anad})$$

$$= 0 + 366$$

$$= 366$$



Admissible heuristic and consistent heuristic for optimality of A* search algorithm.

⇒ Heuristic may be:

(1) $h(n) > \text{actual cost}$:

overestimation.

optimal solution can be overlooked

optimal solution is not possible.

(2) $h(n) \geq \text{actual cost}$:

Best case scenario, if $h(n)$ approximates actual cost, search uses minimum of node to the goal.

(3) $h(n) < \text{actual cost}$:

Underestimation

→ admissible heuristic: $h(n) \leq \text{actual cost}$

→ consistent heuristic: $h(n) \leq c(n, n') + h(n')$

→ Show that the 8 puzzle states are divided into two disjoint sets, such that no state in ~~the~~ one set can be transformed into a state in the other set by any number of moves.

Soln: The 8-puzzle consists of 9 positions on a 3x3 grid. The ~~the~~ puzzle starts in a specific configuration, and the goal is to reach the solved configuration by moving the tiles into the blank space.

Inversions and Solvability:

An inversion occurs when a tile with a higher number appears before a tile with a lower number when reading the puzzle config row by row, left to right.

Two Disjoint Sets:

- Set 1 (Even Inversion): All solvable states
- Set 2 (Odd Inversion): All unsolvable state

7	2	4
5		6
8	3	1

Start state

	1	2
3	4	5
6	7	8

Goal state

$$h_1 = \text{no. of displaced} = 8$$

$$h_2 = 3 + 1 + 2 + 2 + 2 + 3 + 3 + 2 = 18$$

$$h(n) = h_1 + h_2 = 8 + 18 = 26$$

The object is to reach a specified goal state. The standard formulation is as follows:

- State: The state description specifies the location of each of the eight tiles and the blank in one of the nine square.
- Initial state: Any state can be designed as the initial state.
- Actions: The simplest formulation define the action as movements of the blank space Left, Right, up or Down.
- Transition model: Given a state and action, this return the resulting action state.

• Goal test: This checks whether the state matches the goal.

• Path cost: Each step costs 1, so path cost is the number of steps in the path.

1 (even inversion): All solvable states
2 (odd inversion): All unsolvable states